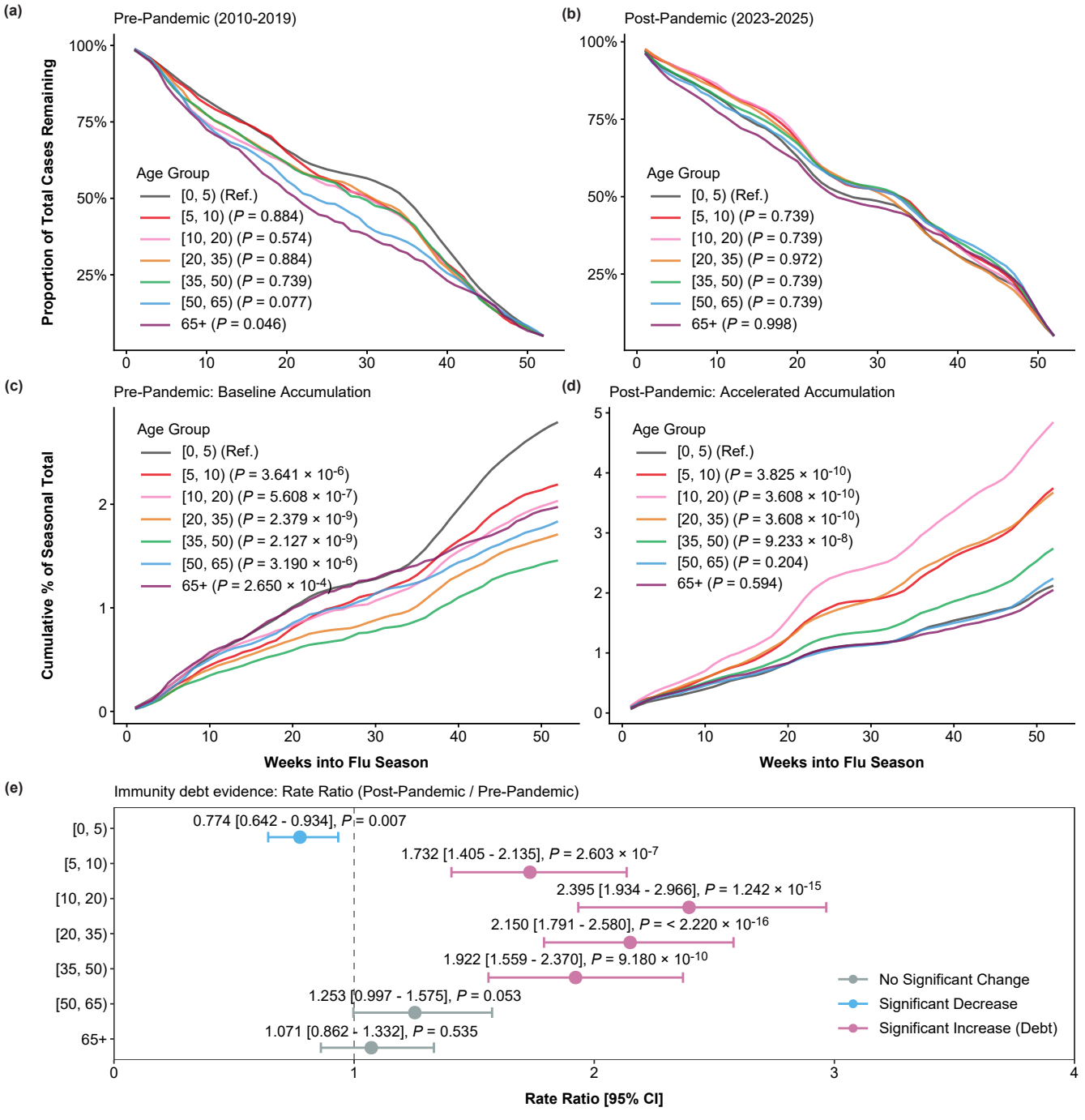

(a)
Pre-Pandemic (2010-2019)
Proportion of Total Cases Remaining
100%
75%
50%
25%
0
10
20
30
40
50
Age Group
[0, 5) (Ref.)
[5, 10) (P = 0.884)
[10, 20) (P = 0.574)
[20, 35) (P = 0.884)
[35, 50) (P = 0.739)
[50, 65) (P = 0.077)
65+ (P = 0.046)
(b)
Post-Pandemic (2023-2025)
Age Group
[0, 5) (Ref.)
[5, 10) (P = 0.739)
[10, 20) (P = 0.739)
[20, 35) (P = 0.972)
[35, 50) (P = 0.739)
[50, 65) (P = 0.739)
65+ (P = 0.998)
(c)
Pre-Pandemic: Baseline Accumulation
Cumulative % of Seasonal Total
Age Group
[0, 5) (Ref.)
[5, 10) (P = 3.641 × 10-6)
[10, 20) (P = 5.608 × 10-7)
[20, 35) (P = 2.379 × 10-9)
[35, 50) (P = 2.127 × 10-9)
[50, 65) (P = 3.190 × 10-6)
65+ (P = 2.650 × 10-4)
Weeks into Flu Season
(d)
Post-Pandemic: Accelerated Accumulation
Age Group
[0, 5) (Ref.)
[5, 10) (P = 3.825 × 10-10)
[10, 20) (P = 3.608 × 10-10)
[20, 35) (P = 3.608 × 10-10)
[35, 50) (P = 9.233 × 10-8)
[50, 65) (P = 0.204)
65+ (P = 0.594)
Weeks into Flu Season
(e)
Immunity debt evidence: Rate Ratio (Post-Pandemic / Pre-Pandemic)
[0, 5)
[5, 10)
[10, 20)
[20, 35)
[35, 50)
[50, 65)
65+
0.774 [0.642 - 0.934], P = 0.007
1.732 [1.405 - 2.135], P = 2.603 × 10-7
2.395 [1.934 - 2.966], P = 1.242 × 10-15
2.150 [1.791 - 2.580], P = < 2.220 × 10-16
1.922 [1.559 - 2.370], P = 9.180 × 10-10
1.253 [0.997 - 1.575], P = 0.053
1.071 [0.862 - 1.332], P = 0.535
No Significant Change
Significant Decrease
Significant Increase (Debt)
Rate Ratio [95% CI]